SIEMENS

INSTALLATION INSTRUCTIONS
Model SBGA-34
Audible Base

INTRODUCTION

The Model SBGA-34 Audible Base from Siemens Industry, Inc., consists of a standard Series 11 base combined with an audible device compatible with all H-series, 8700 series, Desigo and Cerberus series addressable detectors. All field wiring terminates at two four-position terminal blocks located on the back of the unit. The SBGA-34 is used with the polarity insensitive HFP-11, HFPT-11, HFPO-11, SFP-11*, SFPT-11* and SFPO-11*, 8710, 8712, 8713, FDO421, FDT421, FDOT421, FDOOT441, FDOOTC441, OP921, OH921, HI921, OOH941, OOHC941 detectors. The SBGA-34 operates as described below (Refer to Figures 1, 2, 3, 4, 5, 6 and 7 for proper wiring connections).

**SFP-11, SFPT-11 and SFPO-11 detectors are approved for use in CANADA only.*

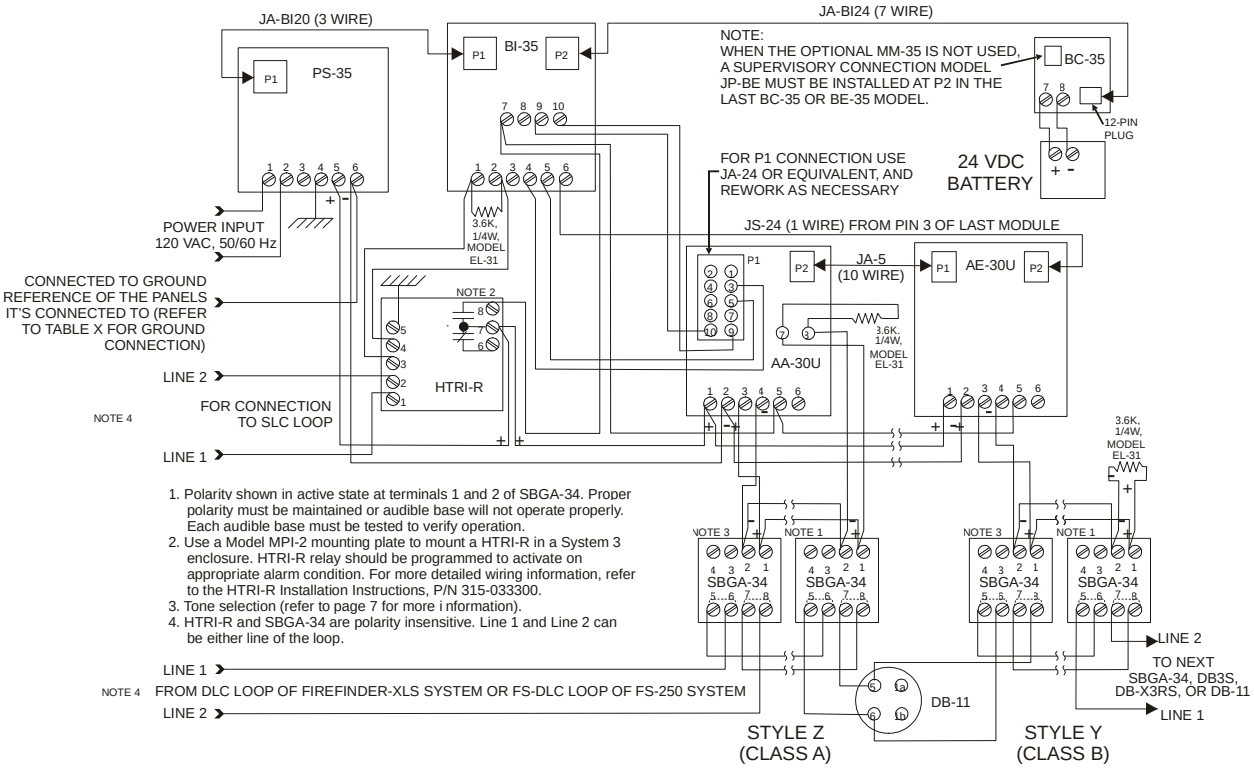
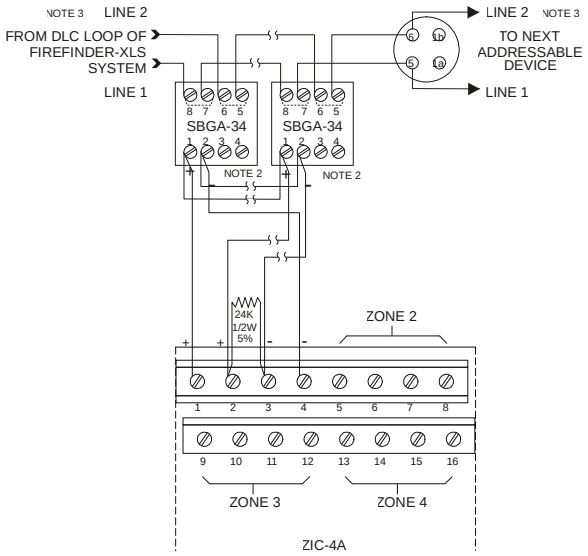


Figure 1
Installation/Wiring Diagram for SBGA-34 Powered by Auxiliary Power Supply with Battery Backup

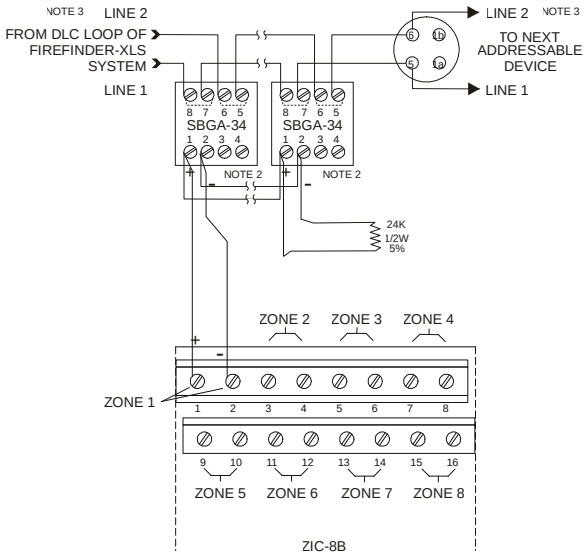
Siemens Industry, Inc.
Building Technologies Division
Florham Park, NJ

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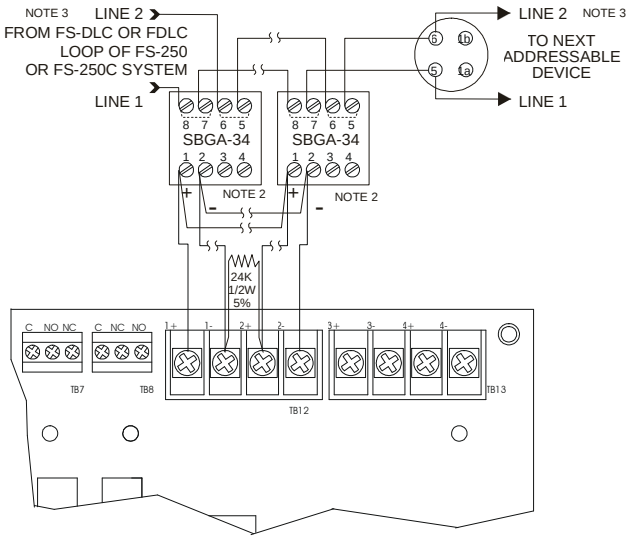
1. Polarity shown in active state at terminals 1 and 2 of SBGA-34 and 1-4 of ZIC-4A. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. SBGA-34 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. SBGA-34 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. ZIC-4A Notification Applicance Circuits provide 4 outputs that can be configured for Class A. or Class B. Refer to the ZIC-4A Installation Instructions, P/N 315-033050.
6. EOL resistor, 24k ohms, ½ watt, 5%, P/N 140-820405.
7. For synchronized operation using the ZIC output, tone selection in the SBGA-34 must be set to Steady.

Figure 2
Installation/Wiring Diagram for SBGA-34 Audible Powered by ZIC-4A (FireFinder-XLS System)



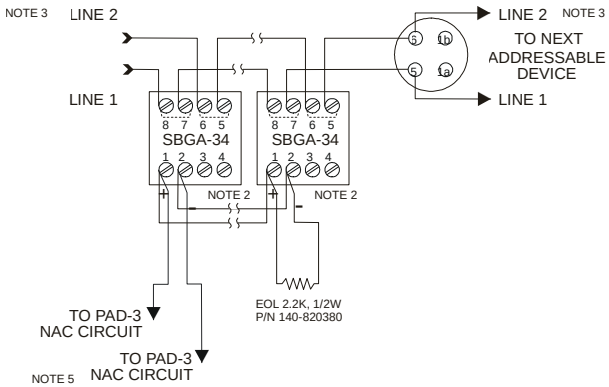
1. Polarity shown in active state at terminals 1 and 2 of SBGA-34 and 1-4 of ZIC-8B. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. SBGA-34 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. SBGA-34 shown as Style Y (Class B).
5. ZIC-8B Notification Applicance Circuits provide 8 outputs that can be configured for Class B. Refer to the ZIC-8B Installation Instructions, P/N 315-048670.
6. EOL resistor, 24k ohms, ½ watt, 5%, P/N 140-820405.
7. For synchronized operation using the ZIC output, tone selection in the SBGA-34 must be set to Steady.

Figure 3
Installation/Wiring Diagram for SBGA-34 Audible Powered by ZIC-8B (FireFinder-XLS System)



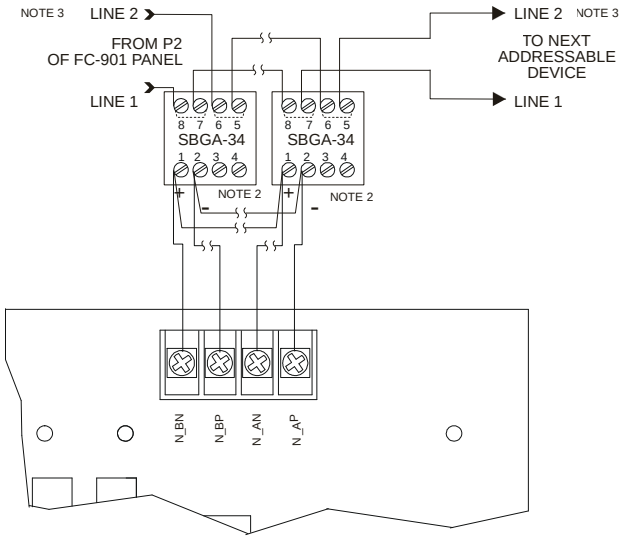
1. Polarity shown in active state at terminals 1 and 2 of SBGA-34 and 1+, 1-, 2+ and 2- of FS-250/FS-250C NAC circuits. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. SBGA-34 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. SBGA-34 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. FS-250/FS-250C Notification Applicance Circuits (TB12 and TB13) provide 4 outputs that can be configured for Class B or two outputs that can be configured for Class A. Refer to the FS-250 Operation, Installation and Maintenance Manual, P/N 315-049353, or the FS-250C Operation, Installation and Maintenance Manual, P/N 315-049589C, as applicable, to set jumpers JP1-4 for Class A or Class B operation.
6. EOL resistor, 24k ohms, ½ watt, 5%, P/N 140-820405.

Figure 4
Installation/Wiring Diagram for SBGA-34 Audible Powered by NAC Circuits (FS-250/FS-250C Systems and MPC6000/MPC7000)



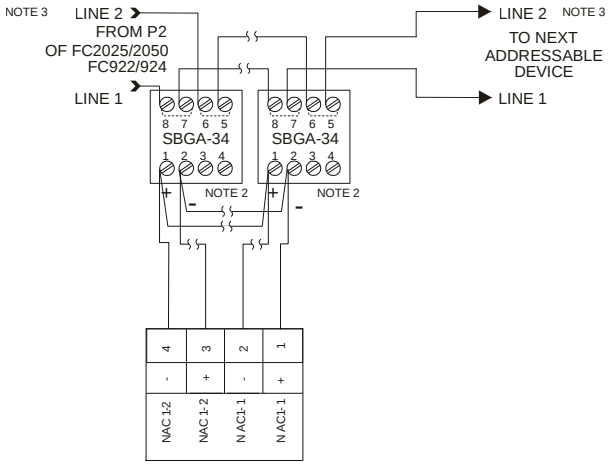
1. NAC circuits polarity shown in active state. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. SBGA-34 is polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. SBGA-34 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. Max. Current 3.0 AMPS. Refer to PAD-3 Installation Instructions, P/N 315-099082.

Figure 5
Installation/Wiring Diagram for SBGA-34 Audible Powered by PAD-3



1. Polarity shown in active state at terminals 1 and 2 of SBGA-34 and 1+, 1-, 2+ and 2- of FC-901 NAC circuits. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. SBGA-34 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. SBGA-34 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. FC901 Notification Appliance outputs that can be configured for Class B or Class A. Refer to the FC-901 Operation, Installation and Maintenance Manual, P/N 315-XXXXXX, as applicable, to set jumpers JP1-4 for Class A or Class B operation.

Figure 6
Installation/Wiring Diagram for SBGA-34 Audible connected to FC901 Panel



1. Polarity shown in active state at terminals 1 and 2 of SBGA-34 and 1+, 1-, 2+ and 2- of FC2025/2050 FC922/924 NAC circuits. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. SBGA-34 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. SBGA-34 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. FC2025/2050 FC922/924 Notification Appliance outputs that can be configured for Class B or Class A. Refer to the FC2025/2050 Operation, Installation and Maintenance Manual, P/N A6V10337045, or the FC922/924 Installation Instructions, P/N A6V10356958, as applicable, to set jumpers JP1-4 for Class A or Class B operation.

Figure 7
Installation/Wiring Diagram for SBGA-34 Audible to FS20 Panel

The SBGA-34 can generate steady, temporal-3 or temporal-4 tone when powered from 24VDC UL listed aux power source or can be configured to follow the NAC tone pattern that is dependent of the panel being used. Refer to the Tone Selection section to configure the units for stand-alone operation.

APPLICATION NOTE

The SBGA-34 is a UL268 and UL464 listed smoke detector supplementary devices which combines detector base and audible device functions. When used in conjunction with the listed compatible Siemens equipment, the SBGA-34 may be used in lieu of single/multiple station smoke alarms to achieve enhanced, system level functionality.

When used to follow the NAC tone pattern, the SBGA-34 must be set to steady tone and the panel it is connected to must be configured accordingly for the desired application. This allows the SBGA-34 to transition from one tone to another (example from temporal 4 to temporal 3) if the proper criteria is met and allows it to be synchronized with the NAC zone that it is connected.

Typical applications are sleeping rooms in hotels, guest suites, dormitories, day-care, residential boards and care facilities as designated in NFPA 72.

Electrical Ratings	
Voltage Rating	17 – 31VDC Special Applications
Current	Supervisory: Loop: 0.6mA External Power Supply: 0mA Alarm: Loop: 2.8mA External Power Supply: 56mA

Audibility Measurements at 10 ft.	
UL464	79dBA
UL268	85dBA
ULCS525	85dBA

CONSTRAINTS

1. SBGA-34 supervisory current is drawn from the detector loop in which their associated detector is connected. This must be included in the determining the battery size of the panel.
2. SBGA-34 alarm current is drawn both from external power supply source and the detector loop in which their associated detector is connected. Care must be taken to verify that this is properly entered during its battery calculation requirements.
3. SBGA-34 audible section can be energized by an auxiliary power supply such as PS-35 with AU-30U (System 3 power supply), PAD-3 / PAD 4 (Auxiliary Power Supply / NAC Extender) or through the NAC circuit. The SBGA-34 can also be powered by a UL listed 24VDC Auxiliary power supply/ Notification Appliance Circuit listed for fire protection signaling use.
4. The maximum number of SBGA-34 is related to the maximum current rating and power line resistance of the auxiliary power supplies applied on the SBGA-34. Refer to the related installation instructions of the auxiliary power supply. It is also related to the maximum current rating of the loop. Refer to the installation instructions mentioned in the fifth constraint.
5. The maximum loop line resistance cannot exceed the maximum line resistance of the loop driver the SBGA-34 is connected to specified in their respective installation instructions. For

the DLC module (XLS panel), refer to the DLC installation instructions, P/N 315-033090. For the FCI2016-U1/FCI2017-U1 module (250pt/500pt Desigo/Cerberus panel), refer to the FS20 Installation Instructions, P/N A6V10315038. For the FC901MB-U1 (50pt Desigo/Cerberus panel), refer to the FC360 Installation Instructions, P/N A6V10336754.

6. T-tapping of the notification appliance circuit is not allowed.
7. When SBGA-34 Is powered from external power supply source, care must be taken to insure that proper ground fault detection is not compromised In typical application where the external power source do not have the capability for ground fault detection, its ground potential must be connected to the ground potential of the FACP to allow for ground fault supervision.
8. A minimum of 14 AWG must be used.
9. Even though no specific spacing is allocated to the detectors used with this base, refer to NFPA Standard 72 National Fire Alarm Code and the installation instructions of the detector being used (see Table 1).

For additional information on the used in an individual system, refer to the installation instructions for that particular module or contact your authorized Siemens Industry, Inc. representative.

DETECTOR PLACEMENT

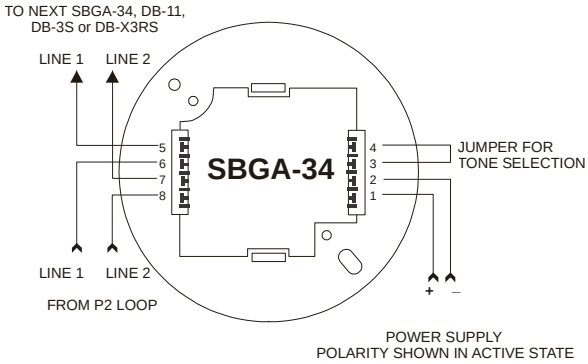
The model SBGA-34 is compatible to the detectors listed in Table 1:

TABLE 1
COMPATIBLE DETECTORS

Detector	Inst. Instr. P/N
HFP-11/HFPT-11	315-033290
HFPO-11	315-034800
SFP-11/SFPT-11*	315-033290C
SFPO-11*	315-033290C
8710, 8712, 8713	315-033290FA
FDO421	A6V10323926
FDOT421	A6V10323934
FDT421	A6V10323930
OP921	A6V10323928
OH921	A6V10323936
HI921	A6V10323932
FDOOT441	A6V10324655
FDOOTC441	A6V10324657
OOHC941	A6V10324661
OOH941	A6V10324659
*SFP-11, SFPT-11 and SFPO-11 detectors are approved for use in Canada only.	

DETECTOR WIRING

Refer to Figure 8. Audible Base Model SBGA-34 should be interconnected as shown in the Installation/Wiring Diagrams as shown in figure 1-7 and wired to the specific system modules and control panels following the appropriate instructions. For operation with Firefinder-XLS, FS250/FS250, FC2025/FC2050, MP6000/7000, FC922/FC924, FC901/FC2005 systems, note any limitations on the number of audible bases permitted on each notification appliance circuit, depending on the total cable resistance for each audible sign circuit (See Electrical Ratings section).



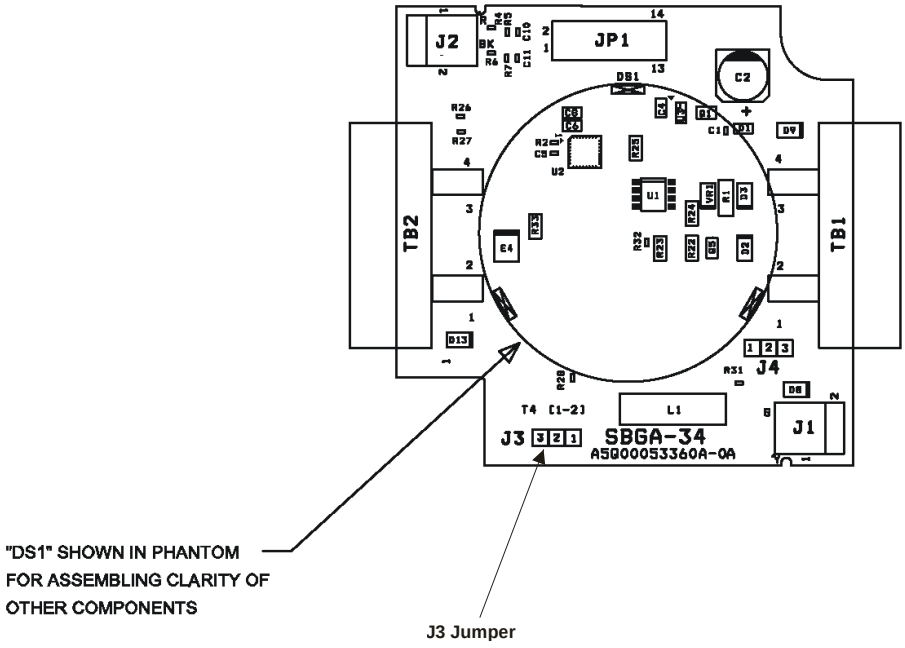
NOTE: Use both Terminals (or leads) for connection, break wire run to provide electrical supervision

Figure 8
SBGA-34 Audible Base Terminal Blocks

TONE SELECTION

With audible base Model SBGA-34, the user can choose either temporal tone (Temporal 3 or Temporal 4) or steady tone (Refer to figure 8).

For Temporal Tone: Terminal 3 and 4 are shorted (insert a jumper) → Default setting is Temporal 3
To Select Temporal 4: Open the unit by unscrewing the DB-11 connected to the base
Move the J3 jumper to 2-3 position and refit the DB-11 to the base.



For Steady Tone: Terminals 3 and 4 are not connected (without a jumper)

NOTE 1: If the Audible Base is intended to follow the NAC coded outputs, the audible base must be set to Steady Tone.

NOTE 2: Temporal 3 signal to be used for evacuation purposes only.

DETECTOR MOUNTING

The detector base comes from the factory prewired. All signaling and initiating circuits are connected directly to the back of the SBGA-34 audible base using the four-position terminal blocks.

NOTE: The Audible Base Model SBGA-34 attaches only to a 4-inch square electrical box with extension.

Mounting the Base

1. Route all wires out from the outlet box.
2. Make connections directly to the audible base terminals located on the back of the base. Refer to the related Installation/Wiring Diagram.
3. After all bases are installed, check loop continuity. For SBGA-34 Detector loop continuity check, refer to the DPU or SDPU Manual, as applicable.
4. If loop continuity is acceptable, proceed with detector head installation.

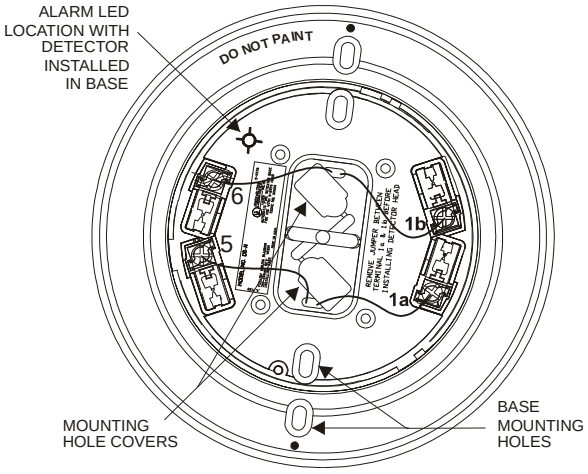


Figure 9
SBGA-34 Top View